

Topoi (ILF)

Goldblatt / Exercises / Reading group

Summer 2026

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Part I

Introduction

Meeting II

Category Theory Pt. 1

1.1 Minimal Exercises

1.1.1 Pre-order Category

For each natural number n , consider the pre-order category $\mathbf{n}$ associated with the pre-order $0 \leq 1 \leq \dots \leq n-1$. How many arrows does $\mathbf{n}$ have?

Solution

For each natural number $k < n$, we can count the number of incoming arrows as

$$|\{j \in \{0, 1, \dots, n-1\} : j \leq k\}| = k + 1.$$

Therefore, there are

$$\sum_{k=0}^{n-1} (k + 1) = \binom{n}{2}$$

arrows in total.

1.1.2 Subcategories of a Monoid

Consider the monoid $\mathbf{N} = (N, +, 0)$ of natural numbers, which gives rise to a category whose collection of objects is the singleton $\{N\}$ and whose collection of arrows is the set N of natural numbers. Which of the following $\mathcal{C}$ are valid subcategories?

- The empty category $\mathcal{C} = \mathbf{0}$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{0\} = \{1, 2, 3, 4, \dots\}$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{1\} = \{0, 2, 3, 4, \dots\}$
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{2\} = \{0, 1, 3, 4, \dots\}$

Solution

- The empty category $\mathcal{C} = \mathbf{0}$

The two conditions for being a subcategory are vacuously true; in fact, the empty category is a

subcategory of every other category. Let $\mathcal{D}$ be any category. Then:

- (i) Every $\mathbf{0}$ -object is a $\mathcal{D}$ -object, vacuously.
- (ii) For every pair of $\mathbf{0}$ -objects a, b , the inclusion $\mathbf{0}(a, b) \subseteq \mathcal{D}(a, b)$ holds, vacuously.

- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{0\} = \{1, 2, 3, 4, \dots\}$

Subcategories must be closed under identities. Since the identity is omitted from the given subquiver, it is not a subcategory.

- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{1\} = \{0, 2, 3, 4, \dots\}$

The given data already form a subquiver. It suffices to check that the subquiver is a category itself with the inherited identity and composition operations. This amounts to checking that the set $N \setminus \{1\}$ is a submonoid of the natural numbers.

1. Since $0 \in N \setminus \{1\}$, it is closed under identities.
2. Let $x, y \in N \setminus \{1\}$. If either $x = 0$ or $y = 0$, then $x + y \in \{x, y\} \subseteq N \setminus \{1\}$. The remaining case is when neither are 0, in which case $x, y \geq 2$. Therefore, $x + y \geq 4$ and so $x + y \in N \setminus \{1\}$.
- $\mathcal{C}$ -objects are $\{N\}$, and $\mathcal{C}$ -arrows are $N \setminus \{2\} = \{0, 1, 3, 4, \dots\}$

The given subquiver is not closed under composition: $1 \in N \setminus \{2\}$ yet $1+1 \notin N \setminus \{2\}$. Therefore, the given subquiver is not a subcategory.

1.2 Exercises for Good Preparation

1.2.1 Endomorphism Monoid

Let $\mathcal{C}$ be a category and X be an object of $\mathcal{C}$. Verify that the collection $\mathcal{C}(X, X)$ of all $\mathcal{C}$ -arrows from X to X forms a monoid with respect to composition.

Solution

We shall check that $(\mathcal{C}(X, X), \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)}, 1_X)$ is a monoid.

Notes

It is worth mentioning that the “hom-collection” $\mathcal{C}(X, X)$ need not be a “small” set whenever $\mathcal{C}$ is not a “locally small” category. In that case, we would call $\mathcal{C}(X, X)$ a “large” (i.e. proper class-sized) monoid.

Let $\mathcal{C}_1$ denote the collection of arrows of $\mathcal{C}$. Notice that $\mathcal{C}(X, X) \times \mathcal{C}(X, X) \subseteq \mathcal{C}_1^{\text{cod} \times \text{dom}} \mathcal{C}_1$

and that $\circ(\mathcal{C}(X, X) \times \mathcal{C}(X, X)) \subseteq \mathcal{C}(X, X)$. Therefore, $\circ$ restricts to a binary operation on the collection $\mathcal{C}(X, X)$.

Notes

It is also worth mentioning that Goldblatt defines composition as an operation with typing $\circ : \mathcal{C}_1^{\text{dom} \times \text{cod}} \mathcal{C}_1 \rightarrow \mathcal{C}_1$, whereas we are using the typing $\circ : \mathcal{C}_1^{\text{cod} \times \text{dom}} \mathcal{C}_1 \rightarrow \mathcal{C}_1$. The two operations are essentially the same, up to swapping arguments.

The associativity of composition in a category translates to associativity of composition on subset $\mathcal{C}(X, X)$. Pedantically, for $f, g, h \in \mathcal{C}(X, X)$:

$$\begin{aligned} & h \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} (g \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} f) \\ &= h \circ (g \circ f) \\ &= (h \circ g) \circ f \\ &= (h \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} g) \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} f \end{aligned}$$

Likewise, $1_X \in \mathcal{C}(X, X)$ acts as an identity for compositions from $\mathcal{C}_1$ that are defined. Therefore, 1_X also acts as an identity on the subset $\mathcal{C}(X, X)$, for which composition is always defined. Pedantically, for $f \in \mathcal{C}(X, X)$:

$$\begin{aligned} 1_X \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} f &= 1_X \circ f = f \\ f \circ|_{\mathcal{C}(X, X) \times \mathcal{C}(X, X)} 1_X &= f \circ 1_X = f \end{aligned}$$

Therefore, $(\mathcal{C}(X, X), \circ, 1_X)$ is a monoid (modulo the fact that the collection $\mathcal{C}(X, X)$ can be a proper class).

1.2.2 Sum Category

Let $\mathcal{C}$ and $\mathcal{D}$ be categories. The sum category, denoted $\mathcal{C} + \mathcal{D}$, is a category whose collection of objects is the disjoint union of $\mathcal{C}$ -objects and $\mathcal{D}$ -objects, and whose collection of arrows is the disjoint union of $\mathcal{C}$ -arrows and $\mathcal{D}$ -arrows.

- Describe how to inherit domains, codomains, identity, and composition from both $\mathcal{C}$ and $\mathcal{D}$. Verify that $\mathcal{C} + \mathcal{D}$ is a category.
- Verify that there are no $(\mathcal{C} + \mathcal{D})$ -arrows between any object that came from $\mathcal{C}$ and any object that came from $\mathcal{D}$.
- Find an example of $\mathcal{C}$ and $\mathcal{D}$ where if set union was used instead of disjoint union, then the resulting construction would not yield a category.

Notes

For collections A and B , let $A + B$ denote the disjoint union. To be pedantic:

$$A + B = \{\langle a, 0 \rangle : a \in A\} \cup \{\langle b, 1 \rangle : b \in B\}$$

If either A or B is a “large” set (i.e. a proper class), then $A + B$ is also a “large” set. Also note that $A + B$ is “small” whenever A and B are both “small”.

For operations $f : A \rightarrow C$ and $g : B \rightarrow D$ between collections, let $f + g : A + B \rightarrow C + D$ denote the operation defined as follows:

$$(f + g)(\langle a, 0 \rangle) = \langle f(a), 0 \rangle$$

$$(f + g)(\langle b, 1 \rangle) = \langle f(b), 1 \rangle$$

Solution Part I

Let $\mathcal{C} = (O, A, \text{dom}, \text{cod}, \circ, 1)$ and $\mathcal{D} = (O', A', \text{dom}', \text{cod}', \circ', 1')$ be the two categories. The typing of our operations is as follows:

$$\text{dom} : A \rightarrow O$$

$$\text{cod} : A \rightarrow O$$

$$\circ : A_{\text{dom} \times \text{cod}} A \rightarrow A$$

$$1 : O \rightarrow A$$

$$\text{dom}' : A' \rightarrow O'$$

$$\text{cod}' : A' \rightarrow O'$$

$$\circ' : A'_{\text{dom}' \times \text{cod}'} A' \rightarrow A'$$

$$1' : O' \rightarrow A'$$

(Here, we choose the typing of $\circ$ and $\circ'$ to follow Goldblatt’s convention: $\circ(\langle g, f \rangle) = g \circ f$ whenever $f : a \rightarrow b$ and $g : b \rightarrow c$.)

As a first approximation, we wish to claim that $\mathcal{C} + \mathcal{D} = (O + O', A + A', \text{dom} + \text{dom}', \text{cod} + \text{cod}', \circ + \circ', 1 + 1')$ is a valid category. The proposed operations would have the following typing:

$$\text{dom} + \text{dom}' : A + A' \rightarrow O + O'$$

$$\text{cod} + \text{cod}' : A + A' \rightarrow O + O'$$

$$\circ + \circ' : (A_{\text{dom} \times \text{cod}} A) + (A'_{\text{dom}' \times \text{cod}'} A') \rightarrow A + A'$$

$$1 + 1' : O + O' \rightarrow A + A'$$

This is close to what we need, but our proposed composition does not have the required typing. To be pedantic, we can adjust the domain of $\circ + \circ'$ by precomposing with the following (bijective) operation:

$$(A + A')_{\text{dom} + \text{dom}' \times \text{cod} + \text{cod}'} (A + A') \xrightarrow{\phi} (A_{\text{dom} \times \text{cod}} A) + (A'_{\text{dom}' \times \text{cod}'} A')$$

$$\langle \langle g, 0 \rangle, \langle f, 0 \rangle \rangle \longmapsto \langle \langle g, f \rangle, 0 \rangle \quad \text{for } a \xrightarrow{f} b \xrightarrow{g} c \text{ in } \mathcal{C}$$

$$\langle \langle g', 1 \rangle, \langle f', 1 \rangle \rangle \longmapsto \langle \langle g', f' \rangle, 1 \rangle \quad \text{for } a' \xrightarrow{f'} b' \xrightarrow{g'} c' \text{ in } \mathcal{D}$$

To verify that we have specified all possible input combinations in our definition of ϕ , note that for each $h \in A + A'$ and $n \in \{0, 1\}$, we have that $(\text{dom} + \text{dom}')(\langle h, n \rangle) = n = (\text{cod} + \text{cod}')(\langle h, n \rangle)$.

Therefore:

$$\begin{aligned}
 & (A + A')_{\text{dom} + \text{dom}' \times \text{cod} + \text{cod}'} (A + A') \\
 &= \{ \langle \langle h, m \rangle, \langle k, n \rangle \rangle \in (A + A')^2 : (\text{dom} + \text{dom}')(\langle h, m \rangle) = (\text{cod} + \text{cod}')(\langle k, n \rangle) \} \\
 &= \{ \langle \langle h, m \rangle, \langle k, n \rangle \rangle \in (A + A')^2 : m = n \}
 \end{aligned}$$

Now, we claim that $\mathcal{C} + \mathcal{D} = (O + O', A + A', \text{dom} + \text{dom}', \text{cod} + \text{cod}', (\circ + \circ'), 1 + 1')$ satisfies the axioms of a category. We do so as follows:

- Domain of Composition:

$$\begin{aligned}
 & (\text{dom} + \text{dom}')((\circ + \circ')\phi(\langle \langle g, 0 \rangle, \langle f, 0 \rangle \rangle)) \\
 &= (\text{dom} + \text{dom}')((\circ + \circ')(\langle \langle g, f \rangle, 0 \rangle)) \\
 &= (\text{dom} + \text{dom}')(\langle \circ(\langle g, f \rangle), 0 \rangle) \\
 &= \langle \text{dom}(\circ(\langle g, f \rangle)), 0 \rangle \\
 &= \langle \text{dom}(g \circ f), 0 \rangle \\
 &= \langle \text{dom}(f), 0 \rangle \\
 &= (\text{dom} + \text{dom}')(\langle f, 0 \rangle)
 \end{aligned}$$

and by symmetry (with computation omitted):

$$(\text{dom} + \text{dom}')((\circ + \circ')\phi(\langle \langle g', 1 \rangle, \langle f', 1 \rangle \rangle)) = (\text{dom} + \text{dom}')(\langle f', 1 \rangle)$$

- Codomain of Composition:

Computations follow by a symmetric argument to verifying domain of composition. With computations omitted, we have the following:

$$(\text{cod} + \text{cod}')((\circ + \circ')\phi(\langle \langle g, 0 \rangle, \langle f, 0 \rangle \rangle)) = (\text{cod} + \text{cod}')(\langle g, 0 \rangle)$$

and

$$(\text{cod} + \text{cod}')((\circ + \circ')\phi(\langle \langle g', 1 \rangle, \langle f', 1 \rangle \rangle)) = (\text{cod} + \text{cod}')(\langle g', 1 \rangle)$$

- Associativity:

$$\begin{aligned}
 & (\circ + \circ')\phi((\circ + \circ')\phi(\langle \langle h, 0 \rangle, \langle g, 0 \rangle \rangle), \langle f, 0 \rangle) \\
 &= (\circ + \circ')\phi((\circ + \circ')(\langle \langle h, g \rangle, 0 \rangle), \langle f, 0 \rangle) \\
 &= (\circ + \circ')\phi(\langle \circ(\langle h, g \rangle), 0 \rangle, \langle f, 0 \rangle) \\
 &= (\circ + \circ')(\langle \circ(\langle h, g \rangle), f \rangle, 0) \\
 &= \langle \circ(\circ(\langle h, g \rangle), f), 0 \rangle \\
 &= \langle \circ(\langle h \circ g, f \rangle), 0 \rangle \\
 &= \langle (h \circ g) \circ f, 0 \rangle
 \end{aligned}$$

and by symmetry (with computation omitted):

$$(\circ + \circ')\phi(\langle \langle h, 0 \rangle, (\circ + \circ')\phi(\langle \langle g, 0 \rangle, \langle f, 0 \rangle \rangle) \rangle) = \langle h \circ (g \circ f), 0 \rangle$$

Since $(h \circ g) \circ f = h \circ (g \circ f)$ holds in $\mathcal{C}$, it follows that

$$\begin{aligned} & (\circ + \circ')\phi(\langle(\circ + \circ')\phi(\langle\langle h, 0 \rangle, \langle g, 0 \rangle\rangle), \langle f, 0 \rangle\rangle) \\ &= \langle(h \circ g) \circ f, 0\rangle \\ &= \langle h \circ (g \circ f), 0\rangle \\ &= (\circ + \circ')\phi(\langle\langle h, 0 \rangle, (\circ + \circ')\phi(\langle\langle g, 0 \rangle, \langle f, 0 \rangle\rangle)\rangle) \end{aligned}$$

and symmetrically (with computation omitted) for

$$\begin{aligned} & (\circ + \circ')\phi(\langle(\circ + \circ')\phi(\langle\langle h', 1 \rangle, \langle g', 1 \rangle\rangle), \langle f', 1 \rangle\rangle) \\ &= \langle(h' \circ g') \circ f', 1\rangle \\ &= \langle h' \circ (g' \circ f'), 1\rangle \\ &= (\circ + \circ')\phi(\langle\langle h', 1 \rangle, (\circ + \circ')\phi(\langle\langle g', 1 \rangle, \langle f', 1 \rangle\rangle)\rangle) \end{aligned}$$

- Domain of identity:

$$\begin{aligned} & (\text{dom} + \text{dom}')((1 + 1')(\langle a, 0 \rangle)) & (\text{dom} + \text{dom}')((1 + 1')(\langle a', 1 \rangle)) \\ &= (\text{dom} + \text{dom}')(\langle 1(a), 0 \rangle) &= (\text{dom} + \text{dom}')(\langle 1'(a'), 1 \rangle) \\ &= \langle \text{dom}(1(a)), 0 \rangle &= \langle \text{dom}'(1'(a')), 1 \rangle \\ &= \langle a, 0 \rangle &= \langle a', 1 \rangle \end{aligned}$$

- Codomain of identity:

$$\begin{aligned} & (\text{cod} + \text{cod}')((1 + 1')(\langle a, 0 \rangle)) & (\text{cod} + \text{cod}')((1 + 1')(\langle a', 1 \rangle)) \\ &= (\text{cod} + \text{cod}')(\langle 1(a), 0 \rangle) &= (\text{cod} + \text{cod}')(\langle 1'(a'), 1 \rangle) \\ &= \langle \text{cod}(1(a)), 0 \rangle &= \langle \text{cod}'(1'(a')), 1 \rangle \\ &= \langle a, 0 \rangle &= \langle a', 1 \rangle \end{aligned}$$

- Right identity:

$$\begin{aligned} & (\circ + \circ')\phi(\langle\langle f, 0 \rangle, (1 + 1')((\text{dom} + \text{dom}')(\langle f, 0 \rangle))\rangle) \\ &= (\circ + \circ')\phi(\langle\langle f, 0 \rangle, (1 + 1')(\langle \text{dom}(f), 0 \rangle)\rangle) \\ &= (\circ + \circ')\phi(\langle\langle f, 0 \rangle, \langle 1(\text{dom}(f)), 0 \rangle\rangle) \\ &= (\circ + \circ')(\langle\langle f, 1(\text{dom}(f)) \rangle, 0 \rangle) \\ &= \langle \circ(\langle f, 1(\text{dom}(f)) \rangle), 0 \rangle \\ &= \langle f, 0 \rangle \end{aligned}$$

and symmetrically (with computation omitted) for

$$(\circ + \circ')\phi(\langle\langle f', 1 \rangle, (1 + 1')((\text{dom} + \text{dom}')(\langle f', 1 \rangle))\rangle) = \langle f', 1 \rangle$$

- Left identity:

Computations follow by a symmetric argument to verifying right identity. With computations omitted, we have the following:

$$(\circ + \circ')\phi(\langle(1 + 1')((\text{dom} + \text{dom}')(\langle f, 0 \rangle)), \langle f, 0 \rangle\rangle) = \langle f, 0 \rangle$$

and

$$(\circ + \circ')\phi(\langle(1 + 1')((\text{dom} + \text{dom}')(\langle f', 1 \rangle)), \langle f', 1 \rangle\rangle) = \langle f', 1 \rangle$$

Solution Part II

This follows from the construction above. If $\langle h, m \rangle$ is composable with $\langle k, n \rangle$, then $m = n$ follows, as shown above.

Solution Part III

Consider the categories $(\{0\}, \{1\}, !, !, !, !)$ and $(\{0\}, \{2\}, !, !, !, !)$ where $!$ is a placeholder for the unique possible function in that slot. Had we used set union instead of disjoint union, we would have obtained a quiver with one vertex 0 and two edges 1 and 2 . Although we would have $1 \circ 1 = 1$ and $2 \circ 2 = 2$, it is unclear how to define $1 \circ 2$ and $2 \circ 1$. In particular, the ϕ used in the construction above would no longer be defined on all inputs.

1.3 Extra Exercise

1.3.1 Quivers

A *quiver* is a directed graph that allows loops and multiple parallel edges. In particular, a quiver Q comprises:

- a collection of things called *Q-vertices*;
- a collection of things called *Q-edges*;
- operations assigning to each *Q-edge* f a *Q-vertex* $s(f)$ (the “source” of f) and a *Q-vertex* $t(f)$ (the “target” of f).

Every category has an underlying quiver by forgetting the composition (and identity): each object is a vertex, each arrow is an edge, $s(f) = \text{dom}(f)$, and $t(f) = \text{cod}(f)$.

Two quivers are considered isomorphic when there is a bijection of vertices and a bijection of edges that preserves sources and targets. Two categories are considered isomorphic when there exists an isomorphism of the underlying quivers that additionally preserves composition and identity.

Find an example of two non-isomorphic categories such that the underlying quivers are isomorphic.

Solution

There are many possible solutions to this exercise. Here, we'll cover the smallest solution:

Let $M = (\mathbb{Z}/2\mathbb{Z}, +, 0)$ and $N = (\mathbb{Z}/2\mathbb{Z}, \cdot, 1)$ be the additive monoid and multiplicative monoid of $\mathbb{Z}/2\mathbb{Z}$, respectively. The categories associated to these monoids each have one object and two arrows. Therefore, their underlying quivers are isomorphic.

The second category satisfies the first-order formula $(\forall f)(f \circ f = f)$ as f varies over all arrows, whereas the first category does not. Therefore, the two categories cannot be isomorphic.

Meeting III

Category Theory Pt. 2

(work in progress)

2.1 Minimal Exercises

2.1.1 Iso Arrows

Exercises from Goldblatt section 3.3: in any category, verify the following:

1. Every identity arrow is iso.

Solution

Let 1_x be an identity arrow for an object x . By the identity axioms of a category, $1_x \circ 1_x = 1_x$ and $1_x \circ 1_x = 1_x$. Therefore, 1_x is an isomorphism with inverse 1_x .

2. If $b \xrightarrow{f} c$ is iso, so is $c \xrightarrow{f^{-1}} b$.

Solution

By definition of iso, $f^{-1} \circ f = 1_b$ and $f \circ f^{-1} = 1_c$. Therefore (upon swapping the order of conjuncts), $f \circ f^{-1} = 1_c$ and $f^{-1} \circ f = 1_b$. Thus, f^{-1} is an iso (with inverse f).

3. $a \xrightarrow{f \circ g} c$ is iso if $b \xrightarrow{f} c$, $a \xrightarrow{g} b$ are, with $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

Solution

By computation:

$$\begin{aligned}
 & (g^{-1} \circ f^{-1}) \circ (f \circ g) & (f \circ g) \circ (g^{-1} \circ f^{-1}) \\
 & = g^{-1} \circ (f^{-1} \circ (f \circ g)) & = f \circ (g \circ (g^{-1} \circ f^{-1})) \\
 & = g^{-1} \circ ((f^{-1} \circ f) \circ g) & = f \circ ((g \circ g^{-1}) \circ f^{-1}) \\
 & = g^{-1} \circ (1_b \circ g) & = f \circ (1_b \circ f^{-1}) \\
 & = g^{-1} \circ g & = f \circ f^{-1} \\
 & = 1_a & = 1_c
 \end{aligned}$$

2.1.2 Terminal Objects

Selected exercises from Goldblatt section 3.6:

1. For any category $\mathcal{C}$, prove that all terminal $\mathcal{C}$ -objects (if any) are isomorphic (to each other).

Solution

Let T and U be terminal objects. Since T is terminal, there is an arrow $l_U : U \rightarrow T$. Since U is terminal, there is an arrow $l'_T : T \rightarrow U$. Therefore, we have the following composites:

$$\begin{aligned} l_U \circ l'_T &: T \rightarrow T \\ l'_T \circ l_U &: U \rightarrow U \end{aligned}$$

Since T is terminal, parallel arrows into T coincide. Therefore, $l_U \circ l'_T = 1_T : T \rightarrow T$. Since U is terminal, parallel arrows into U coincide. Therefore, $l'_T \circ l_U = 1_U : U \rightarrow U$. Thus, l_U and l'_T are each iso (and inverse to each other). Therefore, $T \cong U$.

2. Find terminals in $\mathbf{Set}^2$, $\mathbf{Set}^\rightarrow$, and the poset $\mathbf{n}$.

Solution

- A terminal in $\mathbf{Set}^2$ is $\langle 1, 1 \rangle$. Given any other object $\langle X, Y \rangle$, an arrow is $\langle l_X, l_Y \rangle$. If $\langle f, g \rangle : \langle X, Y \rangle \rightarrow \langle 1, 1 \rangle$ is an arrow, then $f : X \rightarrow 1$ and $g : Y \rightarrow 1$ as arrows in $\mathbf{Set}$. Therefore $f = l_X$ and $g = l_Y$ since parallel arrows into terminals coincide.
- A terminal in $\mathbf{Set}^\rightarrow$ is $1 \xrightarrow{1_1} 1$. Given any other object $f : X \rightarrow Y$, an arrow is $\langle l_X, l_Y \rangle$ since the following square commutes in $\mathbf{Set}$ (being parallel arrows into a terminal):

$$\begin{array}{ccc} X & \xrightarrow{l_X} & 1 \\ f \downarrow & & \downarrow 1_1 \\ Y & \xrightarrow{l_Y} & 1 \end{array}$$

If $\langle f, g \rangle : \langle X, Y \rangle \rightarrow \langle 1, 1 \rangle$ is an arrow, then $f : X \rightarrow 1$ and $g : Y \rightarrow 1$ as arrows in $\mathbf{Set}$. Therefore $f = l_X$ and $g = l_Y$ since parallel arrows into terminals coincide.

- The terminal in the poset $\mathbf{n}$ is $n-1$. Note that this is only possible when $n > 0$, because the empty category cannot have a terminal object.

If x is any object, then $x \leq n-1$ since $n-1$ is the maximum element. Therefore, there exists an arrow into $n-1$ from any other object. Since all parallel pairs coincide in a poset, such an arrow $x \rightarrow n-1$ is unique.

2.1.3 Products

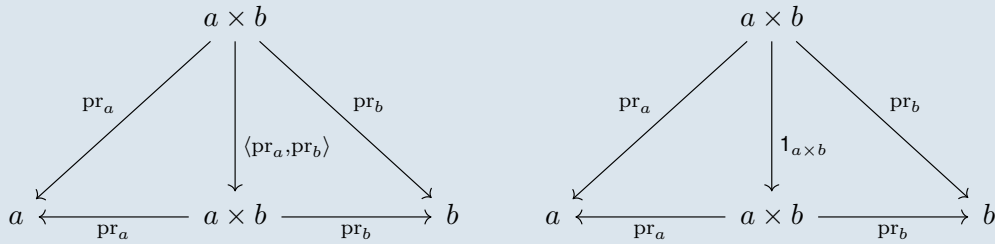
Selected exercises from Goldblatt section 3.8:

Let a, b be two $\mathcal{C}$ -objects that have a product $a \times b$ with projections $\text{pr}_a : a \times b \rightarrow a$ and $\text{pr}_b : a \times b \rightarrow b$.

Exercise 1: Prove that $\langle \text{pr}_a, \text{pr}_b \rangle = 1_{a \times b}$.

Solution

The following diagrams are commutative:



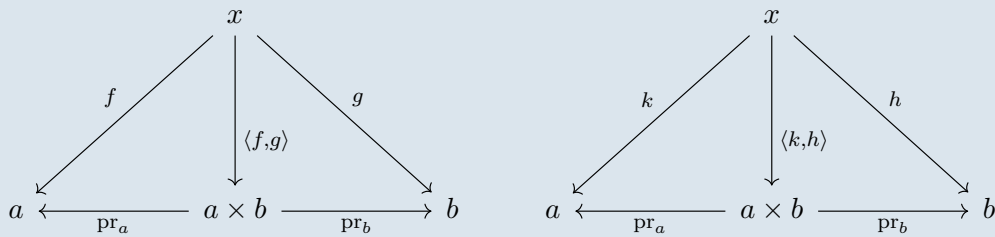
The first diagram commutes by definition of the comparison morphism $\langle \text{pr}_a, \text{pr}_b \rangle$. The second diagram commutes by the right identity axiom.

By uniqueness of cone morphisms into a product, it follows that $\langle \text{pr}_a, \text{pr}_b \rangle = 1_{a \times b}$.

Exercise 2: Prove that if $\langle f, g \rangle = \langle k, h \rangle$, then $f = k$ and $g = h$.

Solution

By definition of the comparison morphism into a product cone, the following diagrams are commutative:

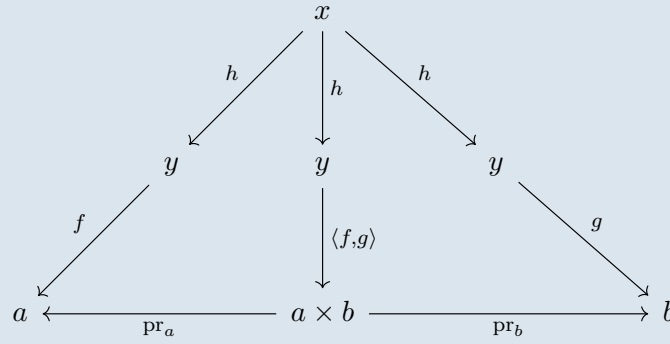


Therefore, $f = \text{pr}_a \circ \langle f, g \rangle = \text{pr}_a \circ \langle k, h \rangle = k$ and $h = \text{pr}_b \circ \langle f, g \rangle = \text{pr}_b \circ \langle k, h \rangle = h$

Exercise 3: Prove that $\langle f \circ h, g \circ h \rangle = \langle f, g \rangle \circ h$.

Solution

Consider the following diagram:



This diagram commutes because

$$\text{pr}_a \circ (\langle f, g \rangle \circ h) = (\text{pr}_a \circ \langle f, g \rangle) \circ h = f \circ h$$

$$\text{pr}_b \circ (\langle f, g \rangle \circ h) = (\text{pr}_b \circ \langle f, g \rangle) \circ h = g \circ h$$

By uniqueness of the comparison arrow, it follows that $\langle f \circ h, g \circ h \rangle = \langle f, g \rangle \circ h$.

Exercise 4: Show that if $\mathcal{C}$ has a terminal object and products, then for any $\mathcal{C}$ -object a , $a \cong a \times 1$ and indeed $\langle 1_a, l_a \rangle$ is iso.

Solution

Firstly, note by definition of the comparison morphism that $\text{pr}_a \circ \langle 1_a, l_a \rangle = 1_a$. Secondly, we utilize the previous exercise to compute that:

$$\langle 1_a, l_a \rangle \circ \text{pr}_a = \langle 1_a \circ \text{pr}_a, l_a \circ \text{pr}_a \rangle = \langle \text{pr}_a, l_a \circ \text{pr}_a \rangle$$

Notice the following arrows are parallel into a terminal: $l_a \circ \text{pr}_a : a \times 1 \rightarrow 1$ and $\text{pr}_1 : a \times 1 \rightarrow 1$. It follows that $l_a \circ \text{pr}_a = \text{pr}_1$. Combined with exercise 1, we have:

$$\langle 1_a, l_a \rangle \circ \text{pr}_a = \langle 1_a \circ \text{pr}_a, l_a \circ \text{pr}_a \rangle = \langle \text{pr}_a, l_a \circ \text{pr}_a \rangle = \langle \text{pr}_a, \text{pr}_1 \rangle = 1_{a \times 1}$$

It follows that $\langle 1_a, l_a \rangle$ is inverse to pr_a . Therefore, $a \cong a \times 1$ via the iso $\langle 1_a, l_a \rangle$.

Exercise 5: Prove that $1_a \times 1_b = 1_{a \times b}$.

Solution

The following diagram commutes by the left/right identity axioms:

$$\begin{array}{ccccc}
 a & \xleftarrow{\text{pr}_a} & a \times b & \xrightarrow{\text{pr}_b} & b \\
 \downarrow 1_a & & \downarrow 1_{a \times b} & & \downarrow 1_b \\
 a & \xleftarrow{\text{pr}_a} & a \times b & \xrightarrow{\text{pr}_b} & b
 \end{array}$$

Therefore, the middle arrow equals the comparison morphism: $1_{a \times b} = 1_a \times 1_b$.

Exercise 6: Prove that a product $b \times a$ exists and that $a \times b \cong b \times a$.

Solution

Part 1:

Suppose “the” product of a and b exists and that we have a product diagram:

$$a \xleftarrow{\text{pr}_a} a \times b \xrightarrow{\text{pr}_b} b$$

We claim that the following is a product diagram, thereby demonstrating the existence of $b \times a$:

$$b \xleftarrow{\text{pr}_b} a \times b \xrightarrow{\text{pr}_a} a$$

Let $f : x \rightarrow b$ and $g : x \rightarrow a$ be arbitrary arrows. Then, $\langle g, f \rangle : x \rightarrow a \times b$ is an arrow such that $\text{pr}_b \circ \langle g, f \rangle = f$ and $\text{pr}_a \circ \langle g, f \rangle = g$. This shows the existence aspect of the universal property for $b \times a$.

Suppose we have another arrow $h : x \rightarrow a \times b$ with the property that $\text{pr}_b \circ h = f$ and $\text{pr}_a \circ h = g$. Then (by swapping order of conjuncts), $\text{pr}_a \circ h = f$ and $\text{pr}_b \circ h = g$.

It follows that $h = \langle g, f \rangle$ by utilizing the uniqueness aspect of the universal property of $a \times b$. This shows the uniqueness aspect of the universal property for $b \times a$.

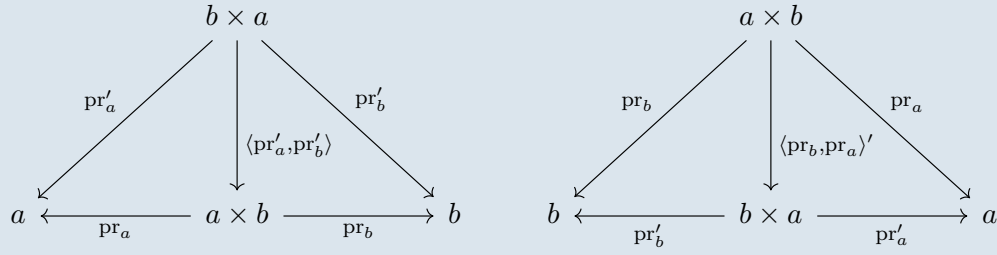
Therefore, $b \times a$ exists with the proposed product diagram.

Part 2:

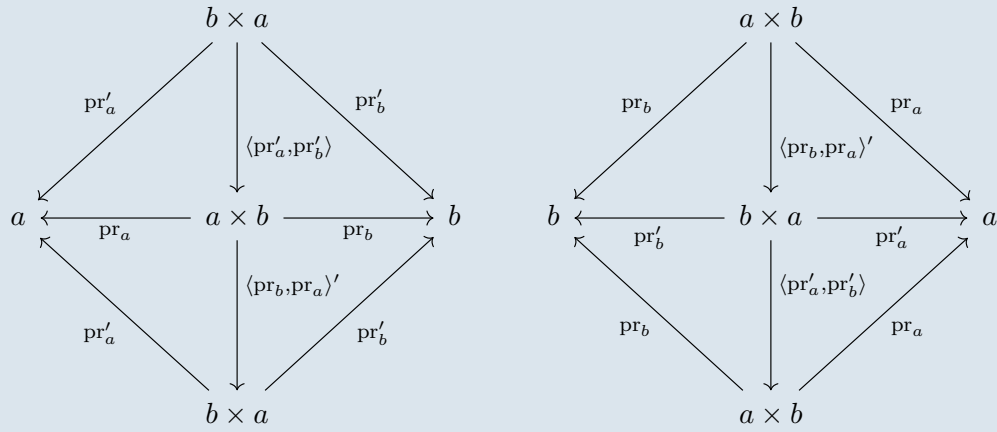
Suppose that we have any other product diagram

$$b \xleftarrow{\text{pr}'_b} b \times a \xrightarrow{\text{pr}'_a} a$$

We would like to show that $b \times a \cong a \times b$ in a way that is compatible with the product diagrams. Consider the following comparison arrows:



By “pasting” together the above commutative diagrams in a compatible manner (e.g. by horizontally flipping one of them and placing it below the other one), we have the following commutative diagrams:



From the uniqueness aspect of the universal property of $b \times a$, the left diagram demonstrates that:

$$\langle \text{pr}_b, \text{pr}_a \rangle' \circ \langle \text{pr}'_a, \text{pr}'_b \rangle = 1_{b \times a}$$

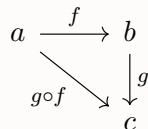
Similarly, from the uniqueness aspect of the universal property of $a \times b$, the right diagram demonstrates that:

$$\langle \text{pr}'_a, \text{pr}'_b \rangle \circ \langle \text{pr}_b, \text{pr}_a \rangle' = 1_{a \times b}$$

2.2 Exercises for Good Preparation

2.2.1 Monic Arrows

Exercises from Goldblatt section 3.1: in any category, consider arrows $a \xrightarrow{f} b$, $b \xrightarrow{g} c$ as in the commutative diagram:



(1) Prove that if f and g are both monic, then $g \circ f$ is monic.

Solution

Let $h, k : x \rightrightarrows a$ be such that $(g \circ f) \circ h = (g \circ f) \circ k$. By associativity and transitivity:

$$g \circ (f \circ h) = (g \circ f) \circ h = (g \circ f) \circ k = g \circ (f \circ k)$$

Since g is monic, it follows that:

$$f \circ h = f \circ k$$

Since f is monic, it follows that:

$$h = k$$

(2) Prove that if $g \circ f$ is monic, then f is monic.

Solution

Let $h, k : x \rightrightarrows a$ be such that $f \circ h = f \circ k$. By associativity and substitution:

$$(g \circ f) \circ h = g \circ (f \circ h) = g \circ (f \circ k) = (g \circ f) \circ k$$

Since $g \circ f$ is monic, it follows that:

$$h = k$$

2.2.2 Terminal Objects

Remaining exercise from Goldblatt section 3.6:

3. Show that an arrow $1 \rightarrow a$ whose domain is a terminal object must be monic.

Solution

Let $f : 1 \rightarrow a$ be an arrow. Let $h, k : x \rightrightarrows 1$ be such that $f \circ h = f \circ k$. Since h and k are already parallel arrows into a terminal object, it follows that $h = k$.

Thus, f is monic.

2.2.3 Isomorphic Objects

Exercise 1 from Goldblatt section 3.4: For any $\mathcal{C}$ -objects a, b, c in any category $\mathcal{C}$, prove the following:

(i) $a \cong a$

Solution

From Goldblatt section 3.3 exercise 1, $1_a : a \rightarrow a$ is an iso, and so witnesses that $a \cong a$.

(ii) if $a \cong b$, then $b \cong a$

Solution

If $a \cong b$ is witnessed by an iso $f : a \rightarrow b$, then by Goldblatt section 3.3 exercise 2, $f^{-1} : b \rightarrow a$ is an iso that witnesses $b \cong a$.

(iii) if $a \cong b$ and $b \cong c$, then $a \cong c$

Solution

If $a \cong b$ is witnessed by an iso $g : a \rightarrow b$ and if $b \cong c$ is witnessed by an iso $f : b \rightarrow c$, then by Goldblatt section 3.3 exercise 3, $f \circ g : a \rightarrow c$ is an iso that witnesses $a \cong c$.

2.2.4 Products

More exercises from Goldblatt section 3.8:

Exercise 7: Show that $(a \times b) \times c \cong a \times (b \times c)$.

Solution

It suffices to show that the following two arrows are inverses of each other:

$$\langle \text{pr}_a \circ \text{pr}_{a \times b}, \langle \text{pr}_b \circ \text{pr}_{a \times b}, \text{pr}_c \rangle \rangle : (a \times b) \times c \rightarrow a \times (b \times c)$$

$$\langle \langle \text{pr}_a, \text{pr}_b \circ \text{pr}_{b \times c} \rangle, \text{pr}_c \circ \text{pr}_{b \times c} \rangle : a \times (b \times c) \rightarrow (a \times b) \times c$$

By computation, we have the following equalities:

$$\text{pr}_a \circ \langle \text{pr}_a \circ \text{pr}_{a \times b}, \langle \text{pr}_b \circ \text{pr}_{a \times b}, \text{pr}_c \rangle \rangle = \text{pr}_a \circ \text{pr}_{a \times b}$$

$$\text{pr}_b \circ \text{pr}_{b \times c} \circ \langle \text{pr}_a \circ \text{pr}_{a \times b}, \langle \text{pr}_b \circ \text{pr}_{a \times b}, \text{pr}_c \rangle \rangle = \text{pr}_b \circ \langle \text{pr}_b \circ \text{pr}_{a \times b}, \text{pr}_c \rangle = \text{pr}_b \circ \text{pr}_{a \times b}$$

$$\text{pr}_c \circ \text{pr}_{b \times c} \circ \langle \text{pr}_a \circ \text{pr}_{a \times b}, \langle \text{pr}_b \circ \text{pr}_{a \times b}, \text{pr}_c \rangle \rangle = \text{pr}_c \circ \langle \text{pr}_b \circ \text{pr}_{a \times b}, \text{pr}_c \rangle = \text{pr}_c$$

By repeated use of Goldblatt exercises 1 and 3 of section 3.8, it follows that:

$$\begin{aligned}
 & \langle \langle \text{pr}_a, \text{pr}_b \circ \text{pr}_{b \times c} \rangle, \text{pr}_c \circ \text{pr}_{b \times c} \rangle \circ \langle \text{pr}_a \circ \text{pr}_{a \times b}, \langle \text{pr}_b \circ \text{pr}_{a \times b}, \text{pr}_c \rangle \rangle \\
 &= \langle \langle \text{pr}_a \circ \text{pr}_{a \times b}, \text{pr}_b \circ \text{pr}_{a \times b} \rangle, \text{pr}_c \rangle \\
 &= \langle \langle \text{pr}_a, \text{pr}_b \rangle \circ \text{pr}_{a \times b}, \text{pr}_c \rangle \\
 &= \langle \mathbf{1}_{a \times b} \circ \text{pr}_{a \times b}, \text{pr}_c \rangle \\
 &= \langle \text{pr}_{a \times b}, \text{pr}_c \rangle \\
 &= \mathbf{1}_{(a \times b) \times c}
 \end{aligned}$$

A symmetric argument (with several computations omitted) would show that:

$$\langle \text{pr}_a \circ \text{pr}_{a \times b}, \langle \text{pr}_b \circ \text{pr}_{a \times b}, \text{pr}_c \rangle \rangle \circ \langle \langle \text{pr}_a, \text{pr}_b \circ \text{pr}_{b \times c} \rangle, \text{pr}_c \circ \text{pr}_{b \times c} \rangle = \mathbf{1}_{a \times (b \times c)}$$

Exercise 8(i): Show that $(f \times h) \circ \langle g, k \rangle = \langle f \circ g, h \circ k \rangle$.

Solution

We may utilize previous exercises to compute as follows, (beginning with the expansion of $f \times g$):

$$\begin{aligned}
 & (f \times h) \circ \langle g, k \rangle \\
 &= \langle f \circ \text{pr}_a, h \circ \text{pr}_c \rangle \circ \langle g, k \rangle \\
 &= \langle f \circ \text{pr}_a \circ \langle g, k \rangle, h \circ \text{pr}_c \circ \langle g, k \rangle \rangle \\
 &= \langle f \circ g, h \circ k \rangle
 \end{aligned}$$

Exercise 8(ii): Show that $(f \times h) \circ (g \times k) = (f \circ g) \times (h \circ k)$.

Solution

We may utilize previous exercises to compute as follows:

$$\begin{aligned}
 & (f \times h) \circ (g \times k) \\
 &= \langle f \circ \text{pr}_a, h \circ \text{pr}_c \rangle \circ \langle g \circ \text{pr}_e, k \circ \text{pr}_{e'} \rangle \\
 &= \langle f \circ \text{pr}_a \circ \langle g \circ \text{pr}_e, k \circ \text{pr}_{e'} \rangle, h \circ \text{pr}_c \circ \langle g \circ \text{pr}_e, k \circ \text{pr}_{e'} \rangle \rangle \\
 &= \langle f \circ g \circ \text{pr}_e, h \circ k \circ \text{pr}_{e'} \rangle \\
 &= (f \circ g) \times (h \circ k)
 \end{aligned}$$

2.3 Extra Exercises

2.3.1 Isomorphic Objects

Exercise 2 from Goldblatt section 3.4: Prove that **Finord** is a skeletal category.

(**Finord** is the full subcategory of **Set** with finite ordinals as objects; refer to Goldblatt section 2.5 example 9 for further details if necessary.)

Solution

Suppose that $a \cong b$ are isomorphic objects of **Finord**. As sets, they have the same cardinality $|a| = n = |b|$ for some finite n . The only object of **Finord** that has exactly n elements is $\{0, 1, \dots, n-1\}$. Therefore, $a = \{0, 1, \dots, n-1\} = b$.

Notes

It's worth noting that even though $a = b$, the isomorphism $a \cong b$ need not be 1_a ; any permutation will do.

2.3.2 Products

Remaining exercises from Goldblatt section 3.8:

Exercise 9: Analyze in detail the formation of the projection arrows $\text{pr}_1^m, \dots, \text{pr}_m^m$, and verify assertions relating to the last diagram (refer to Goldblatt). Show that for any product arrow

$$c \xrightarrow{\langle f_1, \dots, f_m \rangle} a^m,$$

we have $\text{pr}_j^m \circ \langle f_1, \dots, f_m \rangle = f_j$, for $1 \leq j \leq m$.

Solution

For the sake of clarity, let us define inductively that $a^1 = a$ and that for $n \geq 1$, $a^{n+1} = a^n \times a$. For example, our convention means that $a^3 = (a \times a) \times a$.

Notes

The choice $a^{n+1} = a \times a^n$ for $n > 0$ would work as an alternative by symmetry, provided that we consistently adjust all details dependent on this alternative by the same symmetry.

We can define inductively that $\text{pr}_1^1 = 1_a : \underbrace{a^1}_a \rightarrow a$ and that for $n \geq 1$ and $1 \leq j \leq n$,

$$\text{pr}_{n+1}^{n+1} = \text{pr}_a : \underbrace{a^{n+1}}_{a^n \times a} \rightarrow a$$

$$\text{pr}_j^{n+1} = \text{pr}_j^n \circ \text{pr}_{a^n} : \underbrace{a^{n+1}}_{a^n \times a} \rightarrow a$$

Given arrows $f_1, \dots, f_n : c \rightarrow a$, we may also define inductively that $\langle f_1 \rangle = f_1 : c \rightarrow \underbrace{a^1}_a$ and

that for $n \geq 1$,

$$\langle f_1, \dots, f_n, f_{n+1} \rangle = \langle \langle f_1, \dots, f_n \rangle, f_{n+1} \rangle : c \rightarrow \underbrace{a^{n+1}}_{a^n \times a}$$

Showing the universal property of our proposed product diagram $(\text{pr}_j^m)_{1 \leq j \leq n}$ is now an exercise in induction, for which we'll omit the details. Message us on Discord if you'd like the full version!

Showing that $\text{pr}_j^m \circ \langle f_1, \dots, f_m \rangle = f_j$ is also an exercise on induction (on m), which we'll go through now.

Base case $m = 1$:

We only have $j = 1$ to check. By definition,

$$\text{pr}_1^1 \circ \langle f_1 \rangle = 1_a \circ f_1 = f_1$$

Inductive step $m = n + 1$ for $n \geq 1$:

If $j = n + 1$, then

$$\text{pr}_{n+1}^{n+1} \circ \langle f_1, \dots, f_n, f_{n+1} \rangle = \text{pr}_a \circ \langle \langle f_1, \dots, f_n \rangle, f_{n+1} \rangle = f_{n+1}$$

If $j \leq n$,

$$\text{pr}_j^{n+1} \circ \langle f_1, \dots, f_n, f_{n+1} \rangle = \text{pr}_j^n \circ \text{pr}_{a^n} \circ \langle \langle f_1, \dots, f_n \rangle, f_{n+1} \rangle = f_{n+1} = \text{pr}_j^n \circ \langle f_1, \dots, f_n \rangle = f_j$$

where the last step uses the inductive hypothesis.

Exercise 10: Develop the notion of the product $a_1 \times a_2 \times \dots \times a_m$ of m objects (possibly different) and the product of m arrows (possibly different).

Solution

Refer to Meeting III slides for a construction of the product of n (or more objects), possibly all different. If further details are desired, message us on Discord!

2.3.3 Co-products

Exercises from Goldblatt section 3.9:

Exercise 1: Show that $A + B, i_A, i_B$ as just defined (in the category $\mathbf{Set}$, refer to Goldblatt) satisfy the co-product definition.

Solution

Existence of universal property:

Let $f : A \rightarrow C$ and $g : B \rightarrow C$ be two arbitrary functions. Let us define $[f, g] : A + B \rightarrow C$ as follows:

$$[f, g](x) = \begin{cases} f(\text{pr}_A(x)) & \text{if } x \in A \times \{0\} \\ g(\text{pr}_B(x)) & \text{if } x \in B \times \{1\} \end{cases}$$

Note that $A \times \{0\}$ is disjoint from $B \times \{1\}$ since $\langle a, 0 \rangle \neq \langle b, 1 \rangle$ for any combination of $a \in A$ and $b \in B$. Thus, our proposed definition is single-valued.

By computation:

$$\begin{aligned} ([f, g] \circ i_A)(a) &= [f, g](\langle a, 0 \rangle) = f(\text{pr}_A(\langle a, 0 \rangle)) = f(a) & \forall a \in A \\ ([f, g] \circ i_B)(b) &= [f, g](\langle b, 1 \rangle) = g(\text{pr}_B(\langle b, 1 \rangle)) = g(b) & \forall b \in B \end{aligned}$$

This shows the existence aspect of the universal property of a co-product.

Uniqueness of universal property:

Let $h, k : A + B \rightarrow C$ be such that $h \circ i_A = k \circ i_A$ and $h \circ i_B = k \circ i_B$. Let $x \in A + B$. There are two cases to consider:

- If $x \in A \times \{0\}$, then $x = i_A(a)$ for some $a \in A$ because the image of i_A is precisely $A \times \{0\}$. Thus, $h(x) = h(i_A(a)) = k(i_A(a)) = k(x)$
- If $x \in B \times \{1\}$, then $x = i_B(b)$ for some $b \in B$ because the image of i_B is precisely $B \times \{1\}$. Thus, $h(x) = h(i_B(b)) = k(i_B(b)) = k(x)$

Thus, we can conclude that $h = k$.

Exercise 2: If $A \cap B = \emptyset$, show $A \cup B \cong A + B$.

Solution

Let $f : A \hookrightarrow A \cup B$ and $g : B \hookrightarrow A \cup B$ be the two (monic) inclusion arrows. We claim that the morphism $[f, g] : A + B \rightarrow A \cup B$ is an isomorphism. It suffices to check that $[f, g]$ is injective and surjective.

Surjectivity:

Let $x \in A \cup B$. If $x \in A$, then $[f, g](\langle x, 0 \rangle) = f(x) = x$. If $x \in B$, then $[f, g](\langle x, 1 \rangle) = g(x) = x$. Thus, $[f, g]$ is surjective.

Injectivity:

Suppose $\langle x, m \rangle, \langle y, n \rangle \in A + B$ are such that $[f, g](\langle x, m \rangle) = [f, g](\langle y, n \rangle)$. We have four cases

to analyze:

- If $m = 0$ and $n = 1$, then $f(x) = [f, g](\langle x, m \rangle) = [f, g](\langle y, n \rangle) = g(y)$. But since $f(x) \in A$ and $g(y) \in B$, this contradicts $A \cap B = \emptyset$.
- Similarly, if $m = 1$ and $n = 0$, then $g(x) = [f, g](\langle x, m \rangle) = [f, g](\langle y, n \rangle) = f(y)$. But since $g(x) \in B$ and $f(y) \in A$, this contradicts $A \cap B = \emptyset$.
- If $m = n = 0$, then $f(x) = [f, g](\langle x, m \rangle) = [f, g](\langle y, n \rangle) = f(y)$. Since f is injective, it follows that $x = y$.
- If $m = n = 1$, then $g(x) = [f, g](\langle x, m \rangle) = [f, g](\langle y, n \rangle) = g(y)$. Since g is injective, it follows that $x = y$.

In all cases, we can conclude $x = y$. Thus, $[f, g]$ is injective.

Exercise 3: Define the co-product arrow $f + g : a + b \rightarrow c + d$ of arrows $f : a \rightarrow c$ and $g : b \rightarrow d$ and dualise all of the Exercises in section 3.8 (see Goldblatt).

(Write out the dual statement of each exercise; you do not need to write out a dual proof. Though it can feel tedious, this exercise helps develop intuition about dual concepts. It can be insightful to see how a concept and its dual may be fundamentally different constructions, yet behave categorically similar.)

Solution

The arrow $f + g : a + b \rightarrow c + d$ can be analogously defined as:

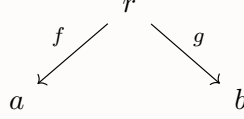
$$f + g = [i_c \circ f, i_d \circ g]$$

Here is the analogous list of properties that various aspects of the co-product satisfy:

- $[i_a, i_b] = 1_{a+b}$
- If $[f, g] = [h, k]$, then $f = h$ and $g = k$
- $[e \circ f, e \circ g] = e \circ [f, g]$
- If $\mathcal{C}$ has an initial object and coproducts, then for any $\mathcal{C}$ -object a , $a \cong a + 0$ and indeed $[1_a, 0_a]$ is iso.
- $1_a + 1_b = 1_{a+b}$
- If $a + b$ exists, then $b + a$ also exists and that $a + b \cong b + a$
- $(a + b) + c \cong a + (b + c)$
- $[f, g] \circ (h + k) = [f \circ h, g \circ k]$
- $(f + h) \circ (g + k) = (f \circ h) + (g \circ k)$

2.3.4 Jointly Monic Pair

Let a, b be $\mathcal{C}$ -objects. Suppose that we have a “span from a to b ”, which is a diagram in $\mathcal{C}$ of the following shape:



We say that $\{f, g\}$ is **jointly monic** if for all parallel pairs $h, k : x \rightarrow r$, the following holds:

$$\text{whenever } f \circ h = f \circ k \text{ and } g \circ h = g \circ k, \text{ then } h = k$$

This is a generalization of being “left-cancellable” in the sense that both conditions need to hold true before we can simultaneously cancel out f and g .

Exercise: Suppose that the product $a \times b$ exists. Prove that $\langle f, g \rangle : r \rightarrow a \times b$ is monic if and only if $\{f, g\}$ is jointly monic.

Solution

Forward direction:

Suppose that $\langle f, g \rangle : r \rightarrow a \times b$ is monic. Let $h, k : x \rightrightarrows r$ be arrows such that $f \circ h = f \circ k$ and $g \circ h = g \circ k$. Then, twice utilizing exercise 3 from Goldblatt section 3.8, we have that:

$$\langle f, g \rangle \circ h = \langle f \circ h, g \circ h \rangle = \langle f \circ k, g \circ k \rangle = \langle f, g \rangle \circ k$$

Since $\langle f, g \rangle$ is monic, it follows that $h = k$.

Reverse direction:

Suppose that $\{f, g\}$ is jointly monic. Let $h, k : x \rightrightarrows r$ be such that $\langle f, g \rangle \circ h = \langle f, g \rangle \circ k$. Utilizing exercise 3 from Goldblatt section 3.8, we have that:

$$\langle f \circ h, g \circ h \rangle = \langle f, g \rangle \circ h = \langle f, g \rangle \circ k = \langle f \circ k, g \circ k \rangle$$

Utilizing exercise 2 from Goldblatt section 3.8, we have $f \circ h = f \circ k$ and $g \circ h = g \circ k$.

Since $\{f, g\}$ is jointly monic, it follows that $h = k$.